

Linear Algebra, Lecture 4

Eigenvalues and Eigenvectors

Autumn term

Outline

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Definition

Let A be an $n \times n$ matrix. An eigenvalue of A is a scalar for which the equation

$$A v = \lambda v$$

has a nonzero solution v . Such a v is an eigenvector of A , and the set of all of them, together with 0 , is the eigenspace.

The characteristic polynomial

The eigenvalues of A are exactly the scalars satisfying

$$\det(A - \lambda I) = 0$$

The left hand side is a polynomial of degree n , called the characteristic polynomial of A . Finding eigenvalues therefore means finding the roots of a polynomial.

Diagonalisation

If A has n linearly independent eigenvectors, collect them as the columns of P . Then

$$P^{-1} A P = D$$

where D is diagonal and holds the eigenvalues.

Powers become cheap: $A^k = P D^k P^{-1}$.

References

Strang, Introduction to Linear Algebra

Axler, Linear Algebra Done Right